

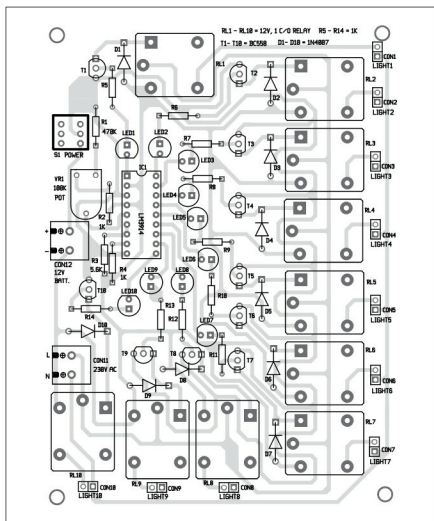


Fig. 3: Component layout of the PCB

varying the voltage at its signal pin 5 through potentiometer VR1. Relays are energised in a progressing manner. Pin 9 is set to 12V positive for operation in bar-graph mode.

Relays energise progressively as voltage at pin 5 increases. This is done by setting VR1, which provides a variable voltage between $\approx 0V$ and $\approx 2V$ at pin 5 of LM3914. Pins 2 and 4 are connected to the ground. Pins 6 and 7 are also connected to the ground through resistor R3. VR1 is rotated from $\approx 0V$ position (all relays remain de-energised) to $\approx 12V$ position (all relays are energised). At other settings of VR1, relays will progressively energise.

The relay driver circuit is built around pnp transistor BC557. Relays are driven to saturation successively as

we turn the control knob of VR1. LED1 through LED10 indicate the particular relays that are energised.

LEDs can be mounted on the panel board along with drawing of the lighting arrangement in the hall. It will help the user to know the on/off status of the lights.

Construction and testing

An actual-size, single-side PCB for effortless hall lights controller is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable plastic box.

Before using the circuit, ensure that supply voltage is connected correctly. ●

The author is an embedded professional and electronics hobbyist

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